

Last-mile Data Analysis and Visualization

Course Project - 2

Submitted in partial fulfillment of the requirements for

Introduction to Multi-Modal Mobility Systems

in

M.Tech Programme

by

Anant Rajput (27252)

Kartikeya Gaur (26371)

Supervisor: **Prof. Vijay Kovvali**



CiSTUP

Indian Institute of Science, Bangalore

Bangalore-560012

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Abstract

The rapid expansion of metro rail networks in Indian cities has highlighted the critical challenge of ensuring efficient and inclusive last mile connectivity. This study investigates last mile travel behavior across three contrasting urban contexts of Delhi, Bangalore, and Nagpur using secondary survey datasets enriched with spatial and demographic information. The methodology involved rigorous data cleaning, preprocessing, and integration with station level metadata, buffer zones, and socio-economic attributes. Descriptive statistics, inferential testing, and spatial analysis were conducted, supported by visualization tools such as Power BI and QGIS. The findings reveal significant gender and income based differences in mode choice, with women showing stronger preferences for affordable and safe modes such as walking and e-rickshaws, while men displayed greater variation in cost tolerance and reliance on private vehicles.

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Chapter 1

Introduction

Urban mobility in India is undergoing a rapid transformation driven by expanding metro systems, growing private vehicle ownership, and the increasing demand for sustainable and inclusive transport solutions. Despite major investments in public transport infrastructure—particularly metro rail networks—many Indian cities continue to face challenges in providing efficient and seamless first- and last-mile connectivity. The effectiveness of a city’s mobility system often depends not only on the quality of its core transit services but also on how well people can access these services from their homes and final destinations.

This study examines three diverse urban contexts to capture the spectrum of challenges in India’s mobility landscape: Delhi, the national capital; Bangalore, a Tier-1 metropolitan and technology hub; and Nagpur, a growing Tier-2 city with emerging transport infrastructure. Delhi faces issues of congestion, overcrowding, and fragmented transport integration despite having the country’s largest metro network. Bangalore struggles with last-mile inefficiencies due to unplanned urban sprawl, heavy traffic, and inadequate pedestrian and cycling infrastructure. Nagpur, while smaller, is at a crucial developmental stage—balancing new metro investments with traditional para-transit modes such as autos and shared vans.

These cities together represent the varying scales, complexities, and stages of urban transport evolution across India. In this context, last-mile data collection and analysis play a critical role in identifying behavioral patterns, accessibility gaps, and system-level inefficiencies. High-resolution, user-level data—covering demographics, mode preferences, travel time, and cost—enables cities to design evidence-based strategies that promote equitable, safe, and sustainable mobility. Understanding the nuances of last-mile choices can help policymakers and planners bridge the gap between public transport availability and actual usability, ensuring that metro systems truly serve as the backbone of urban mobility rather than isolated corridors of movement.

Chapter 2

Objectives

The primary objectives of this study are to:

1. To understand, process, and analyze **mode choice data** for multi-modal travel to investigate **last-mile connectivity** in urban areas.
2. Explore the impact of **user behavior, travel time, costs, and demographic characteristics** on mode choices for last-mile travel.
3. Get hands-on exposure to:
 - **Power BI** for interactive dashboard creation.
 - **QGIS** for spatial mapping and geospatial analysis.
4. Developing informative, interactive, self-explanatory, and user-friendly **dashboards and maps** that can communicate findings effectively to researchers, planners, policymakers, and the public.

Chapter 3

Literature Review

Last-mile transportation, the vital link between mass transit hubs and final destinations, is often inefficient, hindering public transit use and sustainable urban mobility. To address this, data-driven analysis and visualization are crucial for identifying accessibility gaps and designing solutions like shuttle services. Analysis relies on multi-source transportation data, including bicycle-sharing, taxi orders, smart cards, and passive GPS/cell phone data. The process begins with data standardization and last-mile travel demand identification, often defining demand areas using clustering algorithms like DBSCAN within a buffer area of the transit hub.

The analytical insights inform the shuttle service design, which includes bus stop location planning using adaptive clustering (e.g., improved K-means) to minimize walking distance. Route optimization often employs a Genetic Algorithm (GA) to solve the Travelling Salesman Problem (TSP) for cost-effective routes, while scheduling is dynamically adjusted (varied intervals) based on peak and off-peak demand to improve profits and occupancy rates. Furthermore, accessibility measures are utilized, such as gravity-based and cumulative opportunity measures, to quantify current access (e.g., access to jobs by transit) and evaluate the impact of proposed infrastructure improvements, like new pedestrian connections.

A cost-benefit analysis is performed to evaluate the profitability and feasibility of service deployment, acting as a quantitative indicator for planners. The entire analytical process culminates in the creation of informative and user-friendly dashboards and maps (using tools like Power BI and QGIS) to clearly communicate complex findings—such as work accessibility improvements, travel patterns, and station utility—to researchers, policymakers, and the public. This comprehensive data-to-visualization approach helps coordinate decisions across transit operations, road design, and land development to maximize transit system success.

Chapter 4

Methodology

4.1 Data Details and Collection Process

The project uses secondary survey data collected via a questionnaire survey at metro stations in Bangalore, Delhi, and Nagpur. Variables include mode of travel (first-mile, primary, and last-mile connection), reason for mode preference, demographics, household characteristics, and trip characteristics.

4.1.1 Data Cleaning

The raw datasets required systematic cleaning before analysis. First, only the variables relevant to last-mile analysis were retained, such as respondent ID, metro station, last-mile mode, trip purpose, travel time, cost, and socio-demographics. Records with missing, invalid, or zero values in these critical fields were excluded. Duplicate responses, often arising from repeated entries for the same respondent at a given station, were eliminated to maintain data integrity.

Station names posed a major challenge, as the survey data contained spelling variations and inconsistent naming formats across cities (for example, “Rajbagh” versus “Raj Bagh” in Delhi, or “Baiyappanahalli” versus “Baiyyappanahalli” in Bangalore). These inconsistencies were resolved by applying city-specific JSON-based rename maps, which harmonized survey entries with official station names. Finally, outliers in cost, time, and distance values were identified through distributional analysis and corrected or removed where necessary, in order to prevent skewed results.

4.1.2 Data Preprocessing

Once the data was cleaned, it was further preprocessed and enriched through the integration of secondary datasets. The survey responses were merged with station-level geographic information, which included latitude and longitude, to enable spatial mapping. Buffer zone data were incorporated to evaluate accessibility conditions around each station. Mode time summaries were linked with the survey responses to provide an empirical basis for average last-mile travel times across modes. In addition, feature engineering was performed to derive composite measures such as total last mile travel distance and aggregated travel cost per trip. Socio-demographic variables, including gender, age, occupation, and education, were standardized across datasets and recoded into categorical forms suitable for comparative and statistical analysis. Where necessary, categorical

variables were numerically encoded to facilitate inferential testing.

4.2 Data Analysis Steps

4.2.1 Descriptive Analysis

Measures of central tendency (mean, median, mode) and dispersion (standard deviation, variance) were calculated for the combined data for all three cities as shown in Table 4.1.

Table 4.1: Descriptive Statistics Summary

Statistic	Last_Mile_Time	Total_LastMile_Distance_m	Last_Mile_Cost
Min.	1.00	1	0.00
1st Qu.	8.00	400	0.00
Median	10.00	1000	0.00
Mean	13.63	1854	6.63
3rd Qu.	15.00	2000	10.00
Max.	140.00	36000	500.00

4.2.2 Statistical Testing

Chi-square test (Mode choice vs. demographic attributes) is performed as shown in Table 4.2

Table 4.2: Chi-square Test Results

Attribute	χ^2 (Chi-square)	df	p-value	Interpretation
Gender	173.79	74	5.27×10^{-10}	Significant — Mode choice depends on gender
Age	874.95	518	1.00×10^{-20}	Significant — Mode choice depends on age group
Educational qualification	2057.05	616	2.27×10^{-154}	Highly significant — Education strongly influences mode choice
Occupation	3922.80	1332	5.36×10^{-253}	Highly significant — Occupation has strong influence
Monthly household income	1301.91	666	1.62×10^{-43}	Significant — Income affects mode choice
Ownership	611.42	74	1.54×10^{-85}	Highly significant — Vehicle ownership influences mode choice

Chapter 5

Results & Insights

5.1 Key Insights and Results

The comparative analysis across Delhi, Bangalore, and Nagpur revealed important differences and commonalities in last-mile connectivity behavior. The results highlight how socio-demographic characteristics, particularly gender and income, influence mode choice, cost sensitivity, and time trade-offs across the three metro systems.

5.1.1 Gender and Cost Sensitivity

The chord diagram analysis of trip cost by gender showed that women reported a greater share of trips concentrated in lower-cost categories compared to men, particularly in Bangalore and Nagpur. Female respondents consistently opted for affordable options such as walking, shared autos, and e-rickshaws. In contrast, male respondents exhibited a broader spread across cost ranges, with higher utilization of personal vehicles and app-based services (see Figure 5.1). This pattern suggests that cost remains a primary determinant for female commuters, while men demonstrate greater willingness to incur higher last-mile expenses for perceived convenience.

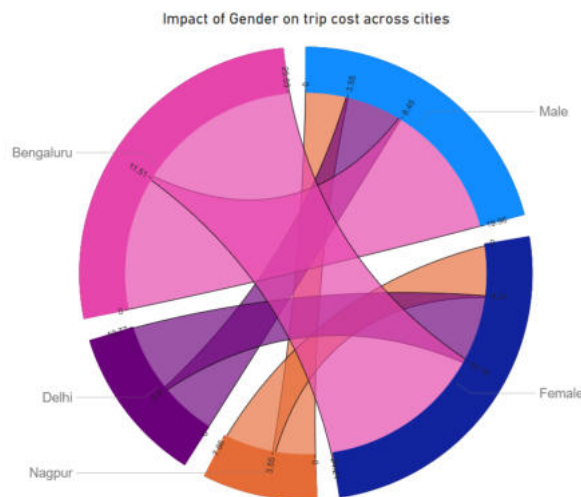


Figure 5.1: Impact of Gender on Trip Cost Across Cities

5.1.2 Gender and Mode Choice

Gender-based differences were also evident in mode choice. Across Delhi and Bangalore, female commuters displayed a stronger preference for modes perceived as safe and predictable, such as walking, shared e-rickshaws, and feeder buses. Male commuters, on the other hand, reported higher use of motorized two-wheelers, app-based taxis, and regular autos. In Delhi, the chord visualization highlighted that e-rickshaws were particularly favored by women, while men exhibited more diversified modal combinations, including mixed modes such as auto-plus-walk or taxi-plus-walk (Figure 5.2). These differences point to the interplay between safety perceptions and affordability in shaping gendered mobility patterns.

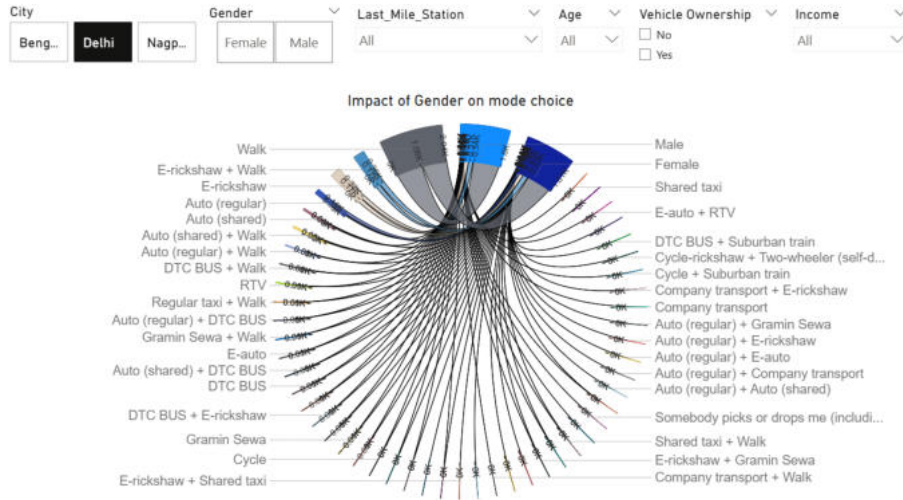


Figure 5.2: Impact of Gender on Mode Choice

5.1.3 Last-Mile Time and Mode Variations

The box plot of last-mile travel times across modes revealed significant heterogeneity (Figure 5.3). Walking consistently showed low variability and short median times, confirming its dominance as a reliable mode for short-distance accessibility. In contrast, auto-rickshaws and app-based taxis exhibited wider variability in travel times, reflecting their dependence on traffic and route conditions. Public modes combined with transfers (e.g., bus plus walk, auto plus bus) had the highest variability, with upper outliers exceeding 60 minutes in Bangalore.

5.1.4 Reasons for Mode Choice

The analysis of primary and secondary reasons behind mode choice provided further explanatory depth. In Bangalore, “safest mode” and “most reliable mode” were cited frequently by women, indicating that non-monetary considerations strongly influence their decisions. Men, however, highlighted speed and convenience more often as primary determinants. In Nagpur, affordability emerged as the overwhelming primary reason, with walking dominating the mode split. The treemap visualizations show the role of affordability and availability (Figures 5.4, 5.5).

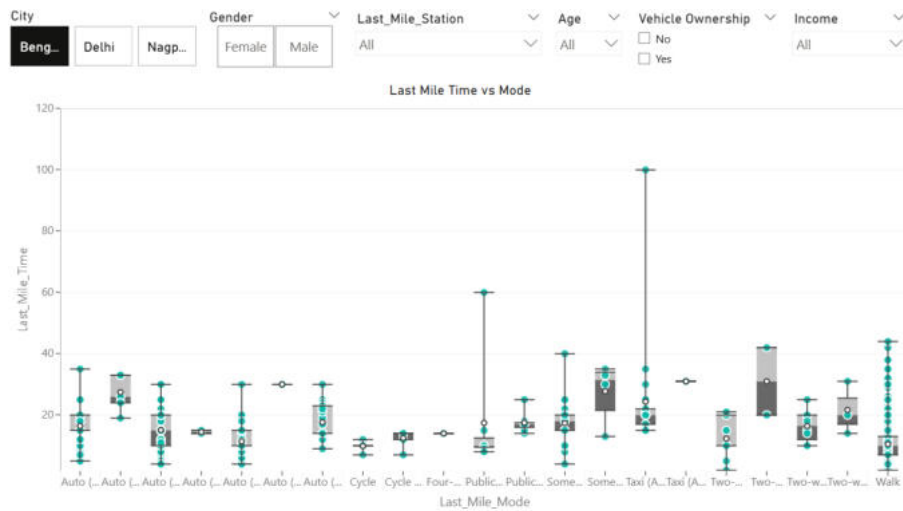


Figure 5.3: Last-Mile Travel Time by Mode

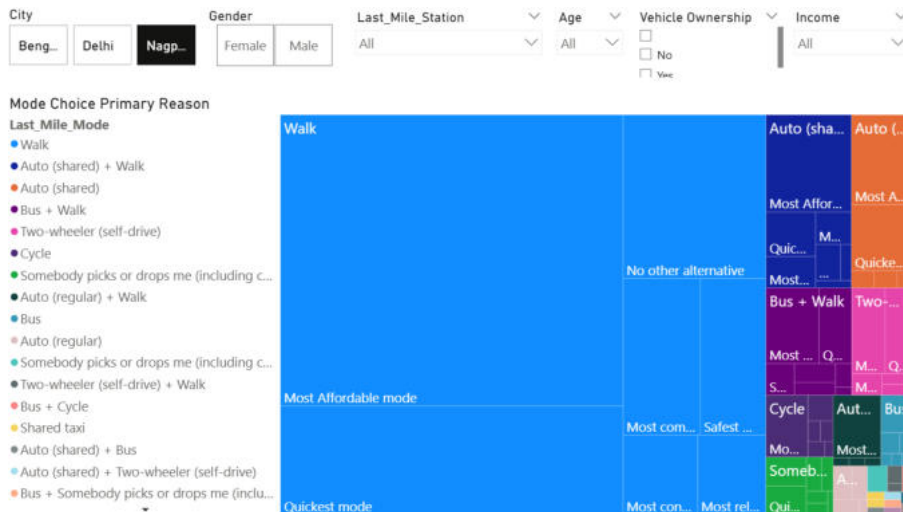


Figure 5.4: Mode Choice Based on Primary Reason

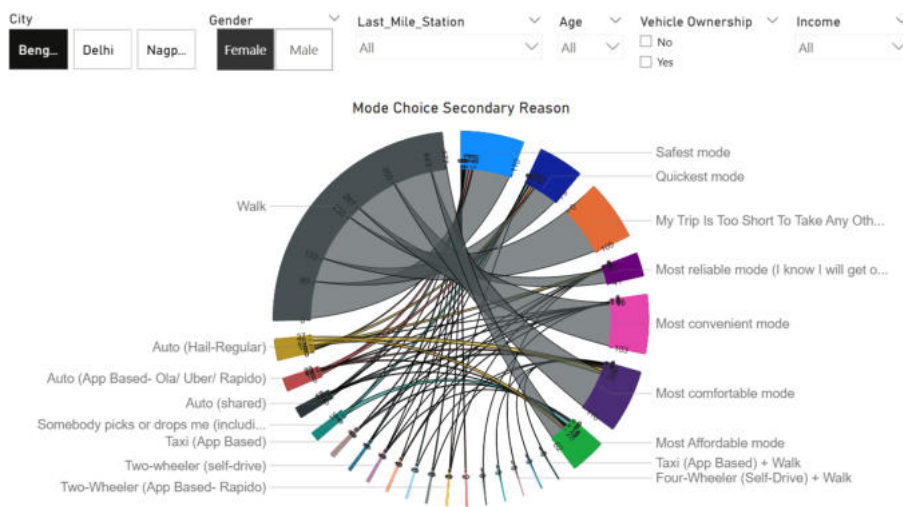


Figure 5.5: Mode Choice Based on Secondary Reason

5.1.5 City-Level Comparisons

Cross-city comparisons highlighted contextual differences. Delhi exhibited the greatest diversity in last-mile modes, supported by a wide para-transit ecosystem of e-rickshaws, shared autos, and feeder buses. Bangalore showed a strong reliance on motorized two-wheelers, taxis, and app-based services, consistent with the city’s higher private vehicle ownership rates. Nagpur, by contrast, was characterized by the dominance of walking. The comparative distributions are shown in Figure 5.6.

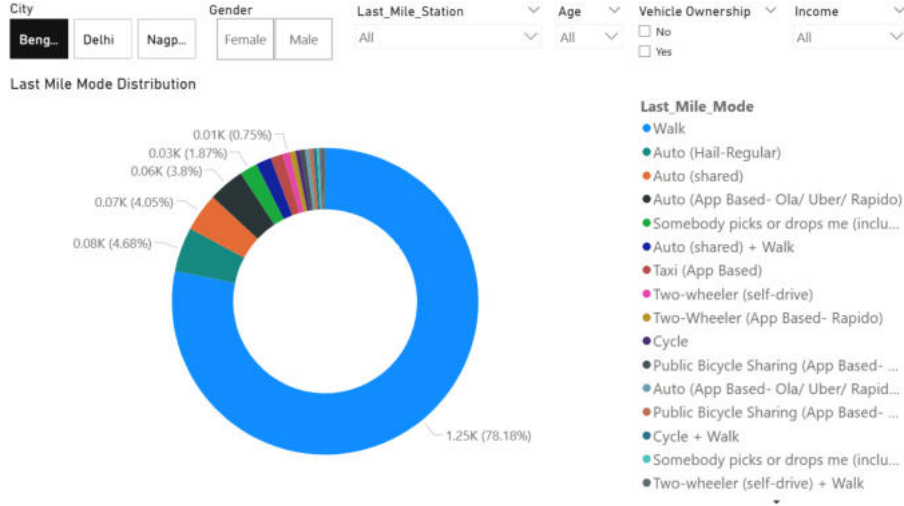


Figure 5.6: Last-Mile Mode Distribution Across Cities

5.1.6 Cost–Time Trade-offs

The scatter plots of last-mile cost versus time further illustrated the trade-offs commuters make (Figure 5.7). Walking and cycling clustered in the low-cost, low-time quadrant, underscoring their efficiency for short distances. Shared autos and e-rickshaws occupied a middle ground, balancing affordability with moderate time requirements. In contrast, app-based taxis and personal four-wheelers were concentrated in the high-cost segment, but without corresponding reductions in time.

5.1.7 Overall Synthesis

While Delhi benefits from a diverse para-transit ecosystem, Bangalore’s reliance on private motorized modes and Nagpur’s dependence on walking highlight the contrasting last-mile landscapes across Indian cities.

Gender emerges as a critical dimension, with women prioritizing safety and affordability, while men exhibit greater variation in mode choice and cost tolerance. These insights suggest that policies to improve last-mile connectivity must be tailored to city-specific contexts, while ensuring inclusive solutions that address the needs of vulnerable groups.

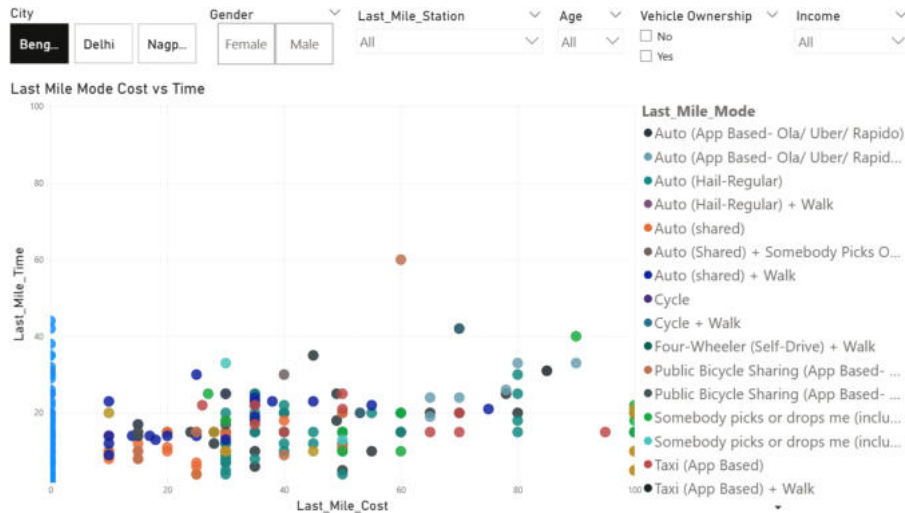


Figure 5.7: Last-Mile Cost vs Time

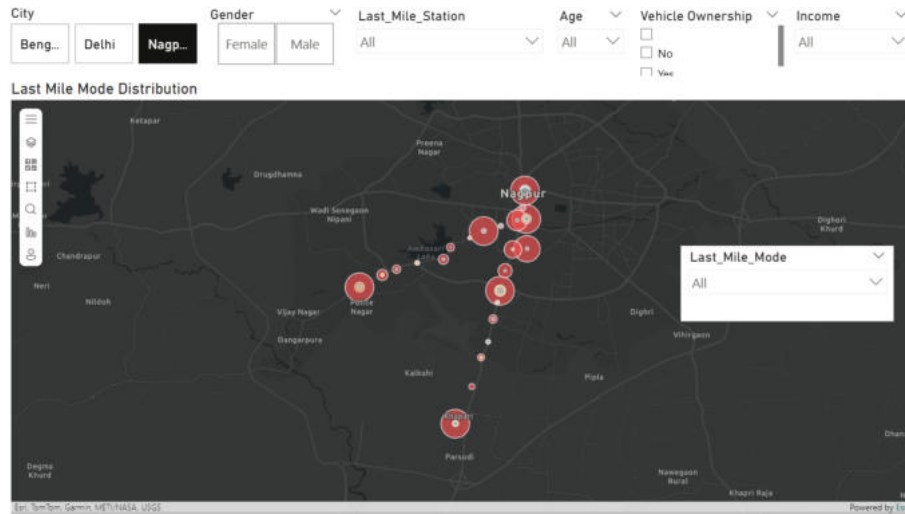


Figure 5.8: ArcGIS Map of Last-Mile Distribution

5.2 Key Insights from Power BI

Insights derived include the most common last-mile modes per city and the impact of gender/income on mode choice.

5.3 Spatial Analysis Findings (QGIS)

Mode Share by Station

- The dominant last-mile mode for Delhi is Walk for all the metro stations as shown in Figure 5.9.
- The dominant last-mile mode for Bangalore is Walk for most of the metro stations as shown in Figure 5.10.
- The dominant last-mile mode for Nagpur is Walk for all the metro stations as shown in Figure 5.11.

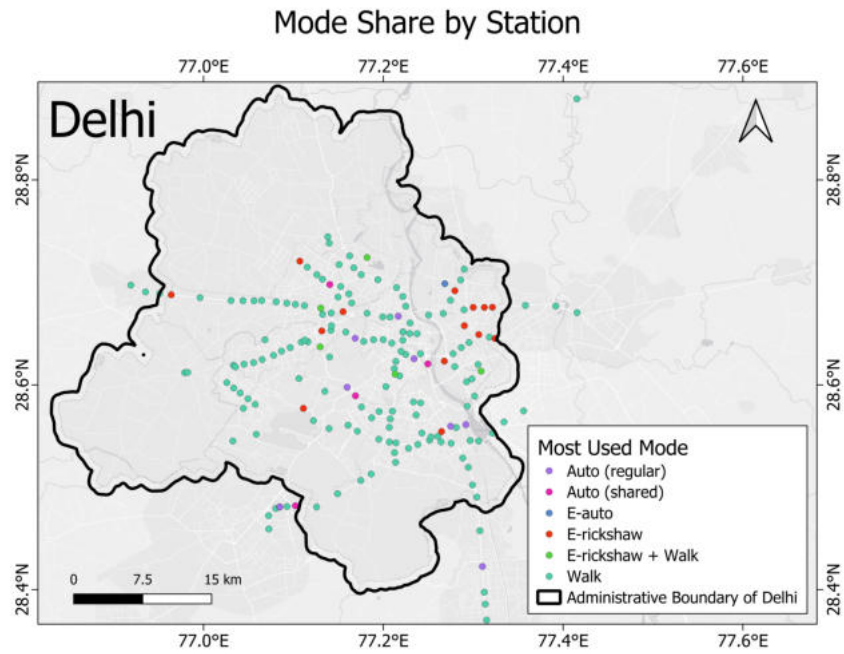


Figure 5.9: Delhi: Most Used Last-Mile Mode Per Station.

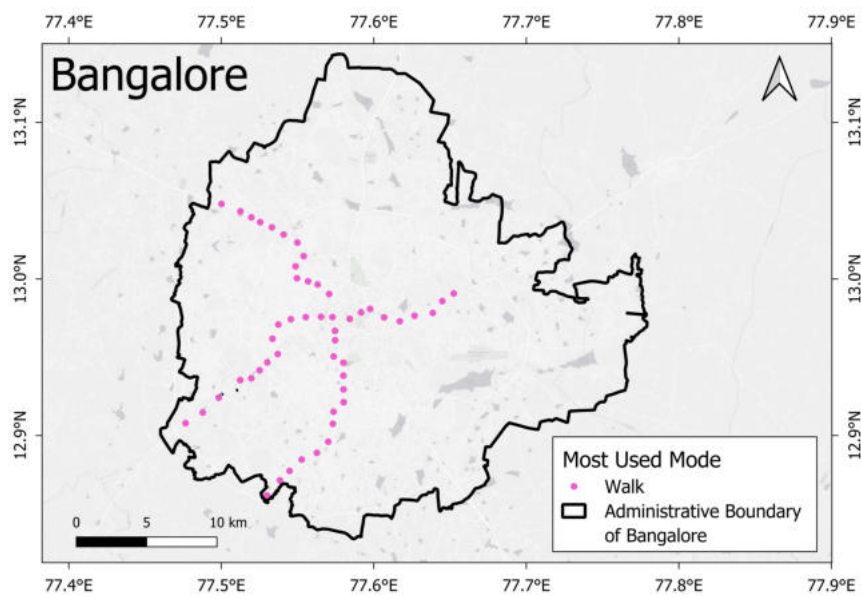


Figure 5.10: Bangalore: Most Used Last-Mile Mode Per Station.

Accessibility Buffers (500m and 1km Percentage Walk Trips)

- In Delhi, the percentage walk trips for 500m buffer have only 5 unique values. Interestingly, 157 out of 192 metro stations have 100% last-mile walk percentage but it is 50 % for Jhilmil and Welcome Metro Stations as shown in Figure 5.12. Whereas for 1000m buffer we have 21 unique values and 152 out of 192 metro stations have 100% last-mile walk percentage. But Mandi House Metro Station have only 18% walk trips as shown in Figure 5.13.
- In Bangalore, the percentage walk trips for 500m buffer have only 2 unique values. Except 1 metro station all other 50 have 100% walk trips as shown in Figure ???. Whereas for 1000m buffer we have 6 unique values and 45 out of 51 metro stations have 100% last-mile walk percentage. But Sri Balagangadaranatha Swamy Hosahalli and Attiguppe Metro Stations have relatively low 83% walk trips as shown in Figure 5.15.
- In Nagpur, the percentage walk trips for 500m buffer have only 3 unique values and 19 out of 23 metro stations have 100% last-mile walk trip as shown in Figure 5.16. Whereas for 1000m buffer we have 6 unique values and 18 out of 23 metro stations have 100% last-mile walk percentage as shown in Figure 5.17.

Travel Time Choropleth (Average Last-Mile Travel Time by Wards)

Choropleth maps for Delhi, Bangalore and Nagpur and shown respectively in Figure 5.18 5.19 5.20

Non-Walk Dependency Zones)

The spatial analysis identifies high non-walk dependency zones and detects the mismatch between accessibility and mode choice.

For Delhi, the non-walk dependency zone is 13.35% by area and contains 38 out of 290 wards. NDMC1 ward is also in this zone which have Connaught Place as shown in Figure 5.21.

For Bangalore, the non-walk dependency zone is 5.74% by area and contains 8 out of 197 wards as shown in Figure 5.22.

For Nagpur, the non-walk dependency zone is 1.87% by area and contains 1 out of 41 wards that is Ward 9 having Kasturchand Park region as shown in Figure 5.23.

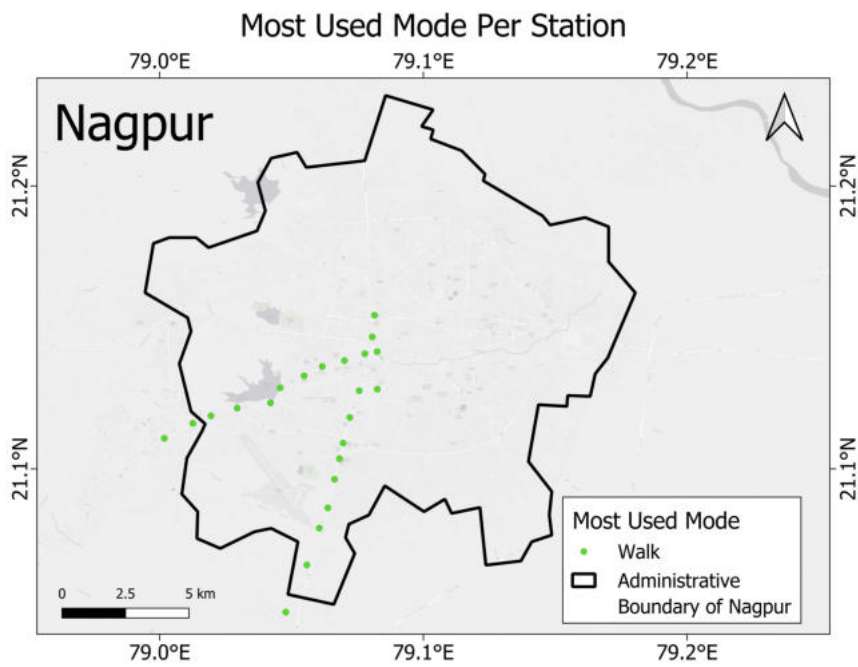


Figure 5.11: Nagpur: Most Used Last-Mile Mode Per Station.

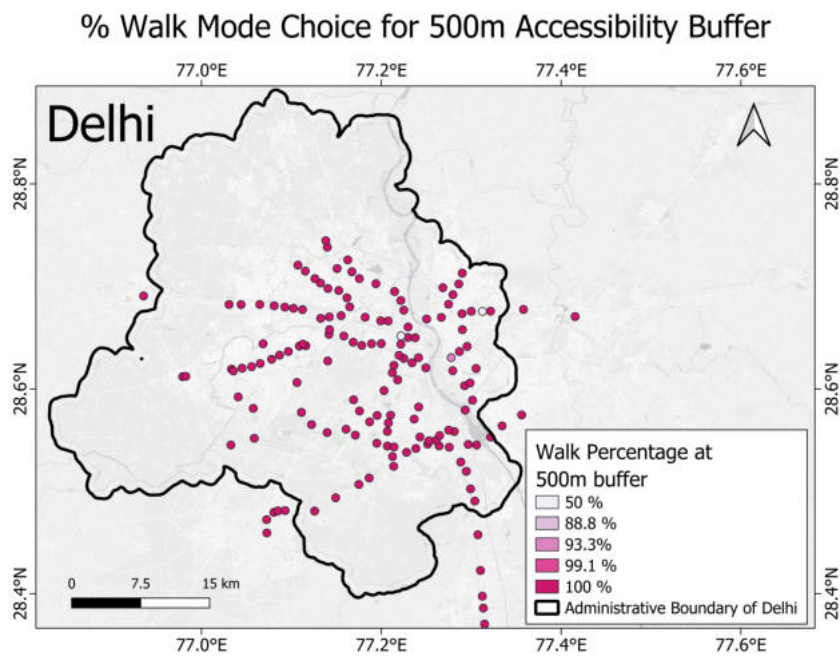


Figure 5.12: Delhi: % Walk Mode Choice for 500m Accessibility Buffer.

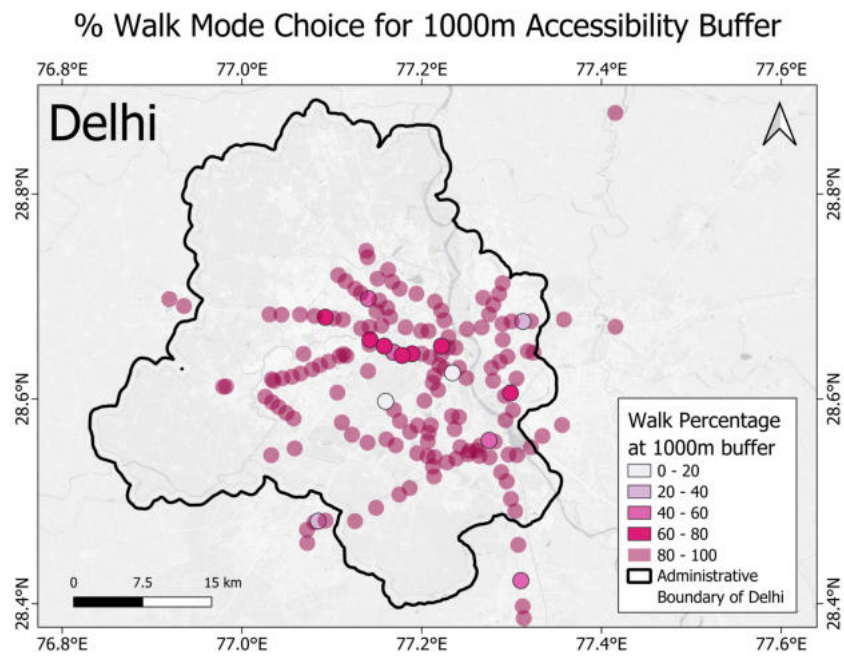


Figure 5.13: Delhi: % Walk Mode Choice for 1000m Accessibility Buffer.

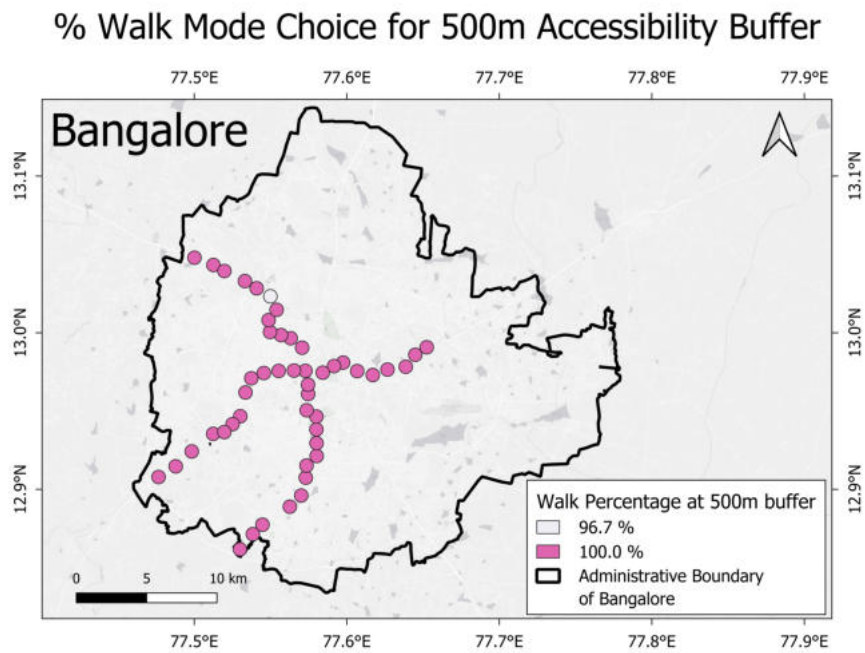


Figure 5.14: Bangalore: % Walk Mode Choice for 500m Accessibility Buffer.

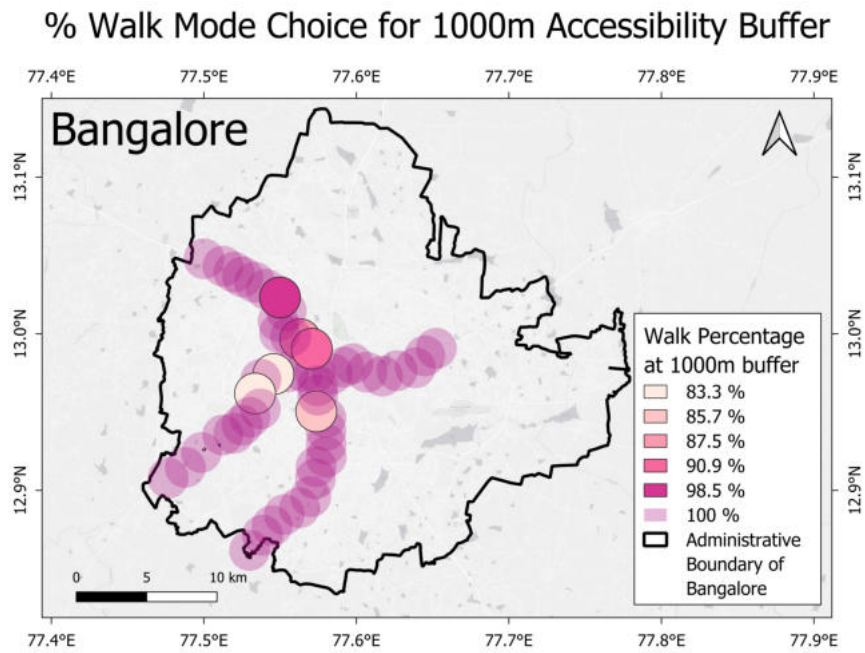


Figure 5.15: Bangalore: % Walk Mode Choice for 1000m Accessibility Buffer.

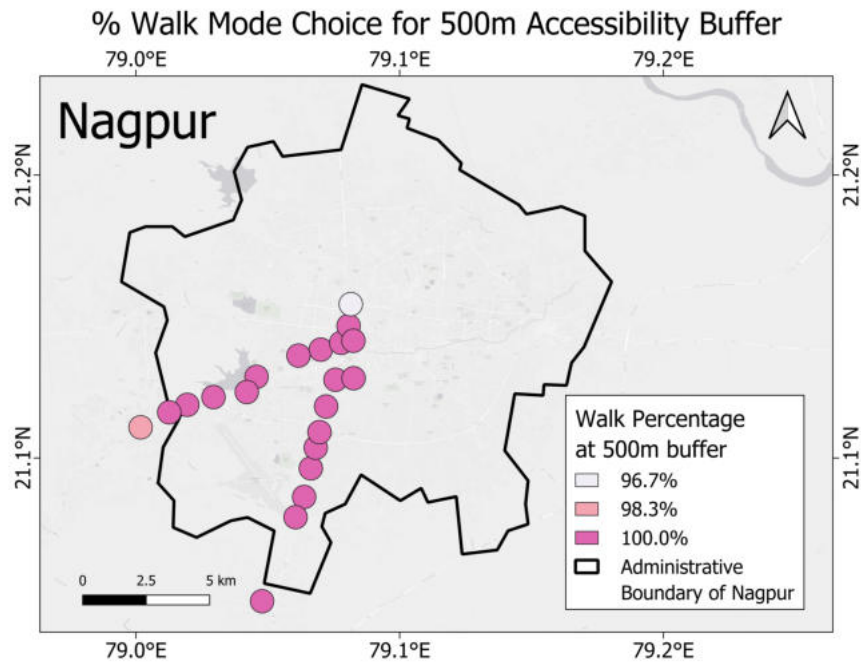


Figure 5.16: Nagpur: % Walk Mode Choice for 500m Accessibility Buffer.

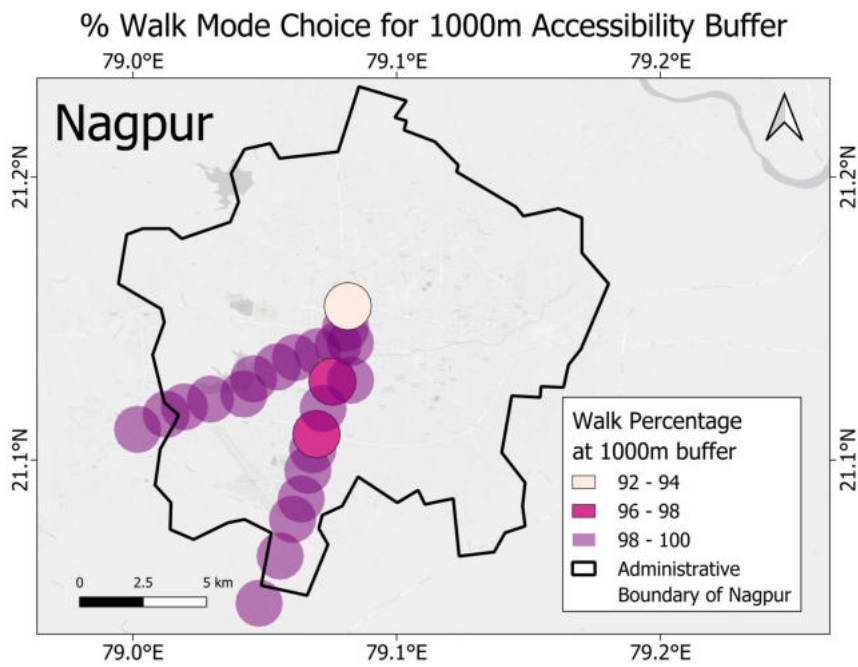


Figure 5.17: Nagpur: % Walk Mode Choice for 1000m Accessibility Buffer.

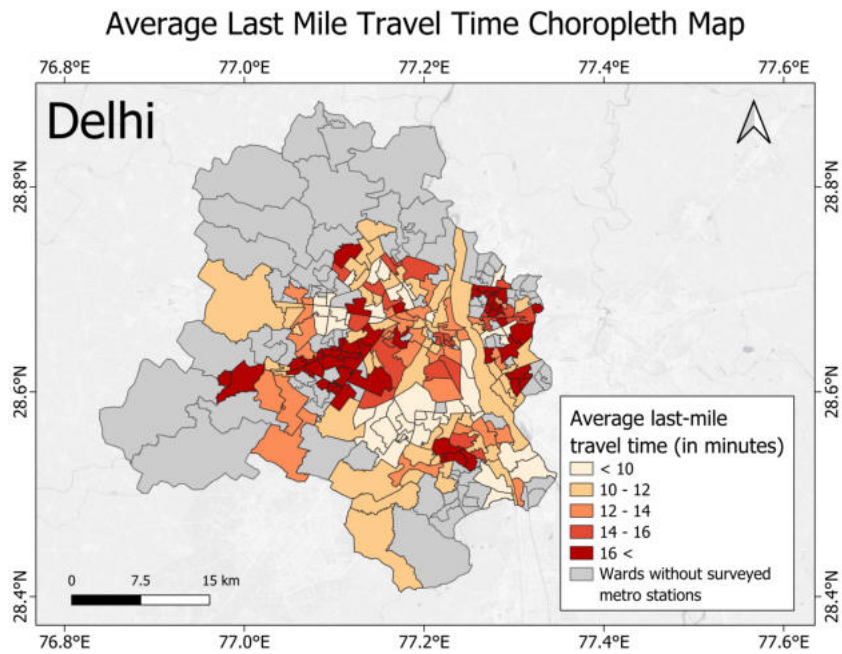


Figure 5.18: Delhi: Average Last-Mile Travel Time Choropleth Map.

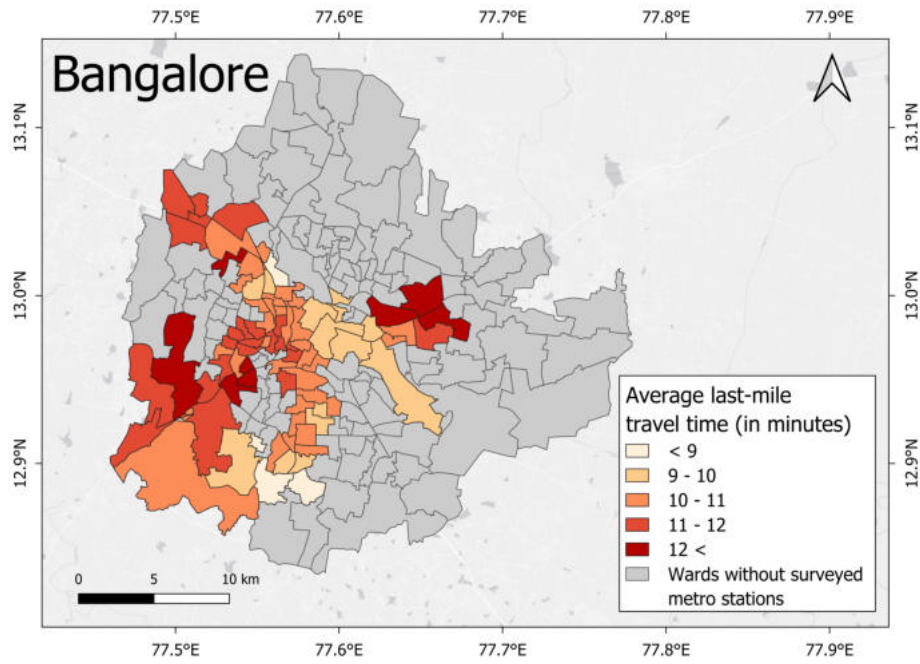


Figure 5.19: Bangalore: Average Last-Mile Travel Time Choropleth Map.

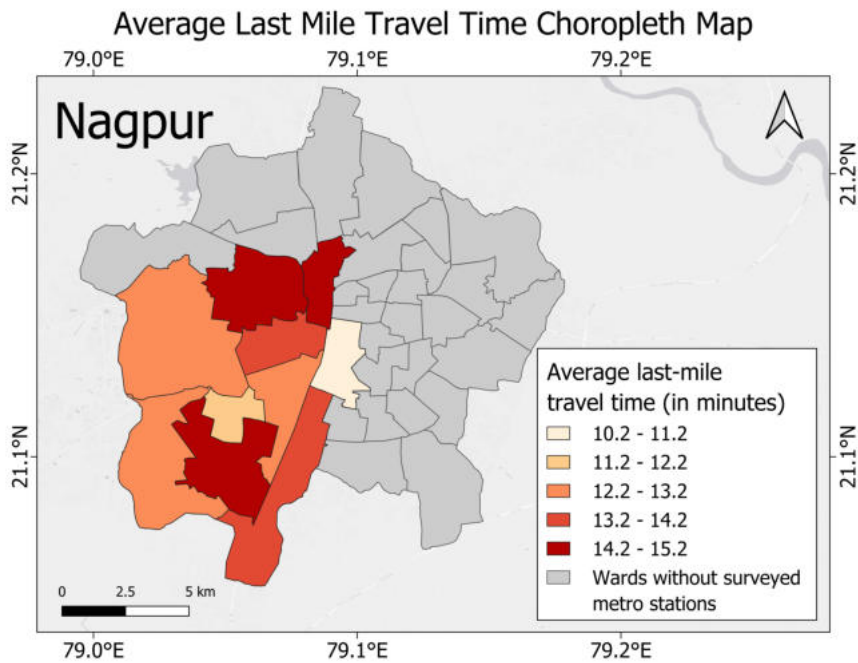


Figure 5.20: Nagpur: Average Last-Mile Travel Time Choropleth Map.

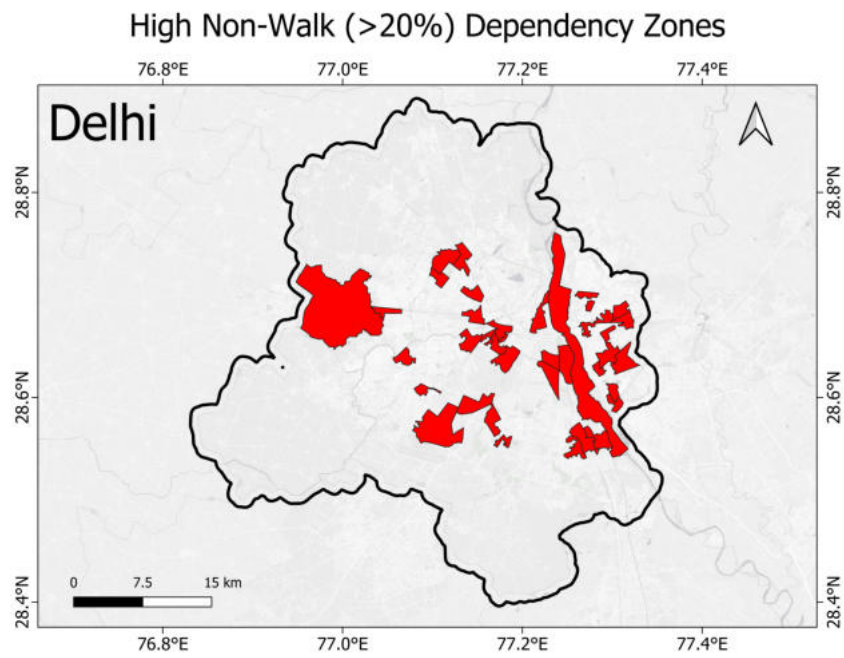


Figure 5.21: Delhi: High Non-Walk Dependency Zones.

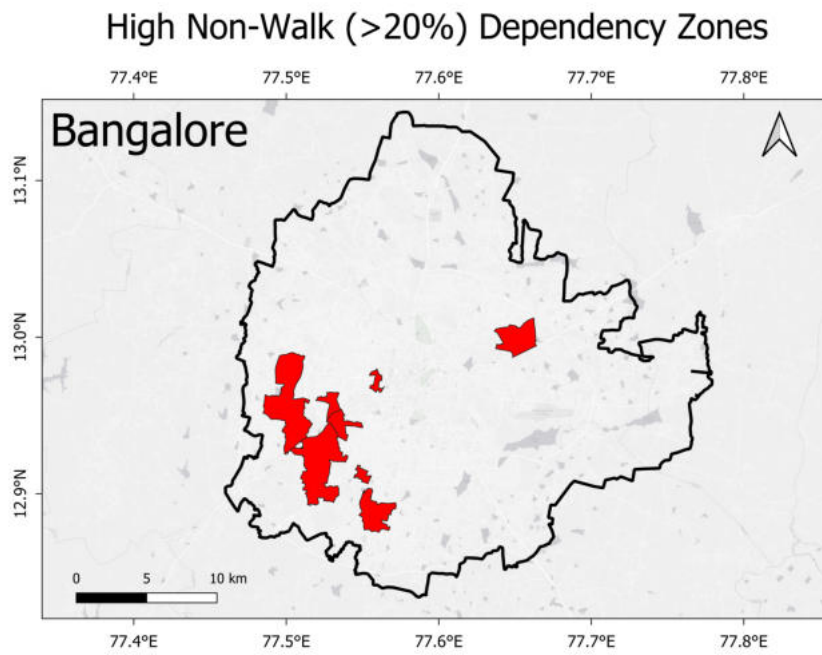


Figure 5.22: Bangalore: High Non-Walk Dependency Zones.

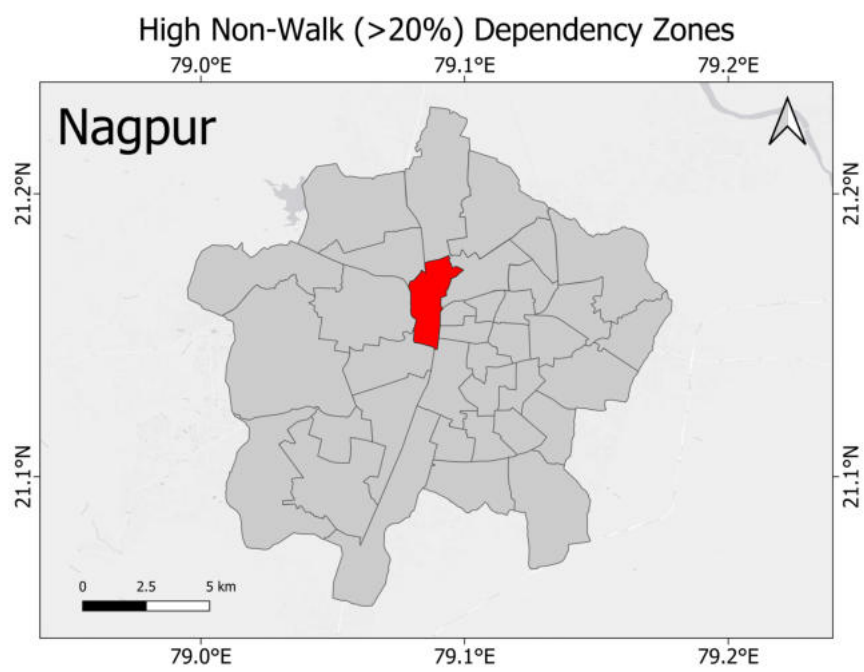


Figure 5.23: Nagpur: High Non-Walk (> 20%) Dependency Zones.

Chapter 6

Critical Discussion

6.1 Key Variables and Mode Preference

The chi-square test results reveal that all the examined socio-economic and demographic attributes — gender, age, educational qualification, occupation, monthly household income, and vehicle ownership — significantly influence the choice of transportation mode. Among these, educational qualification, occupation, and ownership show highly significant relationships, indicating that these factors have the strongest impact on individuals' mode choice behavior.

6.2 City-wise Comparison

A comparison of last-mile connectivity across Bangalore, Delhi, and Nagpur can be derived from percentage of population of the city which can access the facility in 500m radius and last mile mode choice distribution as shown in Table 6.1.

The Figure 6.1 shows that for Delhi it covers only 13.30% population with highest per station 0.25% for Shyam Park which has dominant last-mile mode as E-auto.

The Figure 6.2 shows that for Bangalore it covers only 9.52% population with highest per station 0.44% for Mantri Square and Sri Balagangadaranatha Swamy Hosahalli.

The Figure 6.3 shows that for Nagpur it covers only 9.68% population with highest per station 0.88% for Sitabuldi.

Table 6.1: Last Mile Mode Choice Distribution

Mode	Bangalore	Delhi	Nagpur
Auto	201	217	94
Auto(shared)	0	64	116
E-rickshaw	0	695	0
Other	66	31	51
Walk	1254	2044	1496

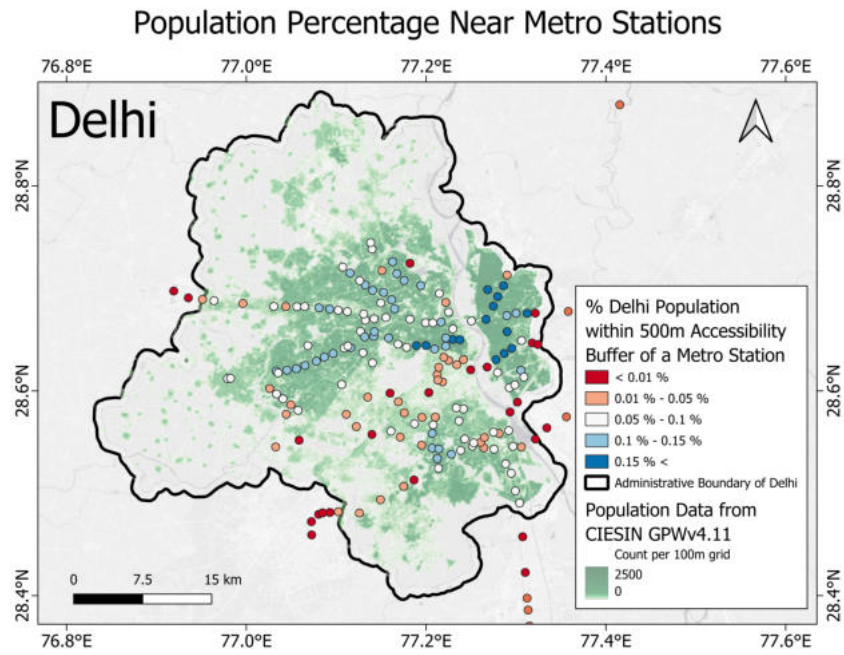


Figure 6.1: Delhi: Population Percentage Near Metro Stations.

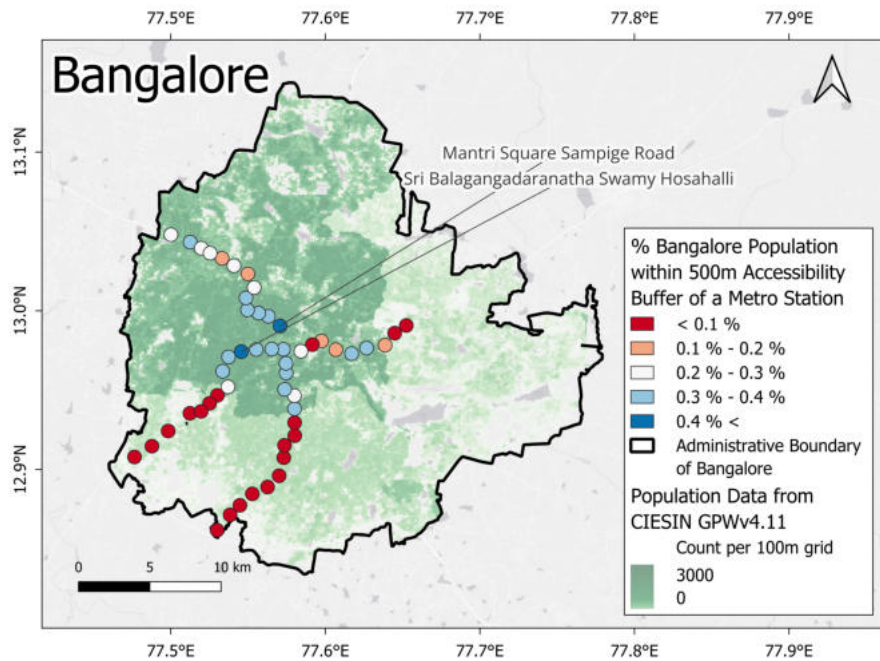


Figure 6.2: Bangalore: Population Percentage Near Metro Stations.

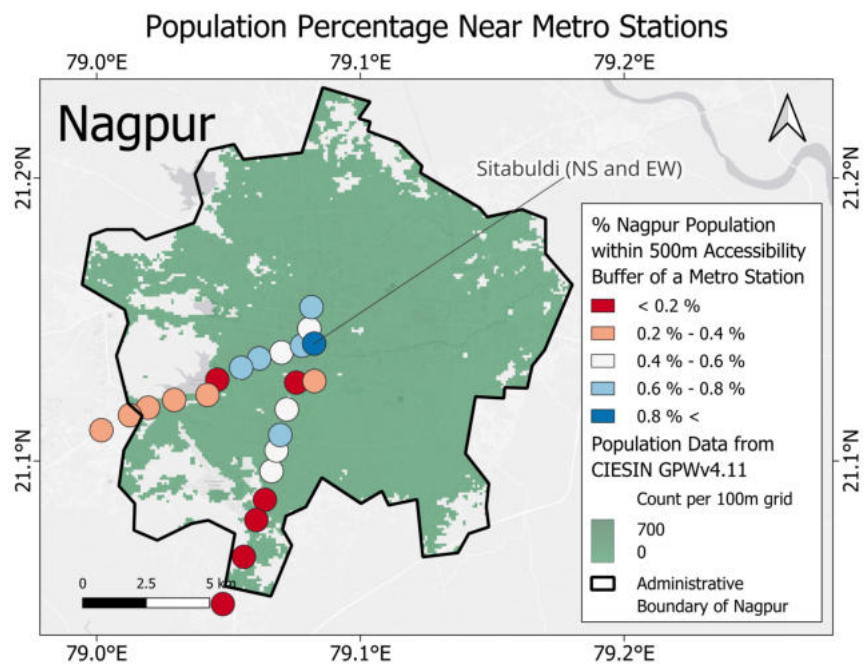


Figure 6.3: Nagpur: Population Percentage Near Metro Stations.

Chapter 7

Recommendations

Improving last-mile connectivity in Indian cities requires an integrated approach that combines infrastructure upgrades, service innovations, and policy interventions. Based on the findings from Delhi, Bangalore, and Nagpur, the following recommendations are proposed:

7.1 General Recommendations

- **Strengthen Non-Motorized Transport (NMT) Infrastructure:** Walking was found to be the dominant last-mile mode across all three cities. Investments in safe pedestrian pathways, wider footpaths, shaded walkways, and bicycle lanes are critical to supporting this demand. Ensuring barrier-free access for persons with disabilities should also be a core priority.
- **Integrate Para-Transit with Metro Systems:** Modes such as e-rickshaws, shared autos, and feeder buses play a crucial role in Delhi and Bangalore. Formalizing and regulating these services through permits, fixed stands, and digital integration with metro smart cards would improve reliability while reducing informal congestion at stations.
- **Adopt Gender-Sensitive Planning:** Women showed a higher preference for safe and predictable modes. Lighting around stations, CCTV coverage, women-only feeder services, and improved last-mile surveillance can enhance safety perceptions and increase public transport adoption among women.
- **Encourage Affordable Shared Mobility:** Cost emerged as a critical determinant, particularly for women and lower-income groups. Shared e-rickshaws, shuttle vans, and app-based pooled rides should be incentivized to ensure affordability without compromising accessibility.
- **Leverage Digital Tools:** Power BI dashboards and QGIS maps developed in this project highlight the usefulness of real-time visualization. Expanding these into operational tools for transport authorities can enable continuous monitoring of accessibility gaps and service utilization.

7.2 Delhi-Specific Recommendations

Delhi exhibits high diversity in last-mile modes, but also a higher share of non-walk dependency zones (13.35%).

- Expand and regulate e-rickshaw and shared auto networks around high-demand stations to reduce crowding and travel uncertainty.
- Improve multimodal integration at key interchange hubs such as Mandi House and Connaught Place by developing multimodal terminals with dedicated feeder bays.
- Introduce fare integration between DMRC metro cards and feeder services to streamline payments and reduce first- and last-mile friction.

7.3 Bangalore-Specific Recommendations

Bangalore relies heavily on private motorized two-wheelers and app-based services, reflecting weak feeder integration.

- Develop a structured feeder bus network operated by BMTC with high-frequency services at metro stations.
- Promote cycle-sharing systems and secure bicycle parking at stations to support shorter last-mile trips in dense residential catchments.
- Implement congestion pricing or parking restrictions around key stations to discourage two-wheeler dominance and shift demand towards shared or public modes.

7.4 Nagpur-Specific Recommendations

Nagpur is at an early stage of metro development, with walking dominating last-mile trips.

- Invest in high-quality pedestrian pathways and safe crossings around all metro stations to strengthen walkability as the backbone of last-mile connectivity.
- Gradually introduce regulated e-rickshaw services for areas beyond the 1 km buffer to extend coverage without creating overdependence on private autos.
- Leverage land-use planning to encourage mixed-use developments around metro corridors, reducing the need for long last-mile trips.

7.5 Policy-Level Recommendations

- Establish a **Unified Metropolitan Transport Authority (UMTA)** in all three cities to coordinate last-mile services across metro, bus, para-transit, and NMT.
- Implement **public-private partnerships (PPPs)** to bring in app-based aggregator models for feeder services, with government regulation ensuring affordability and safety.

- Adopt **performance-based metrics** (e.g., average last-mile time, cost, safety perception scores) as Key Performance Indicators (KPIs) for urban mobility planning.

Chapter 8

Conclusion

The data indicates that Delhi has the highest percentage of non-walk-dependent areas (13.35%) and population coverage within a 500 m accessibility buffer (13.30%), suggesting relatively broader access to public transport infrastructure.

However, the maximum population served per station is lowest in Delhi (0.25%), implying more evenly distributed accessibility. In contrast, Nagpur, with the smallest non-walk-dependent area (1.87%), shows the highest maximum population coverage per station (0.88%), indicating fewer stations but higher concentration of accessibility. Bangalore exhibits moderate values across all parameters.

Overall, the results highlight differing patterns of transport accessibility and spatial distribution among the three cities, with Delhi offering wider but less concentrated accessibility, and Nagpur providing focused accessibility around fewer transit nodes.

Table 8.1: Key Takeaways

City	% Non-walk dependent area	% City Population within 500m accessibility buffer	Max % City Population within 500m accessibility buffer per station
Delhi	13.35	13.30	0.25
Bangalore	5.74	9.52	0.44
Nagpur	1.87	9.68	0.88

City	Female (%)	Male (%)
Bangalore	78.19	78.16
Delhi	61.11	66.33
Nagpur	76.67	76.95

Table 8.2: Percentage of walking by gender in three cities